



JURUTERA SIBER RAWSEC

Smart Healthcare Appointment Management Platform

JURUS Level 1 (Analyst) Cyber Engineering Challenge

Role:	Lead Cyber Engineering Analyst	Duration:	Three (3) Weeks
Focus:	Healthcare Infrastructure Modernisation	Level:	Level 1 Competency

BACKGROUND

The National Community Healthcare Agency has initiated a nationwide digital modernisation programme intended to improve accessibility and efficiency across public healthcare clinics.

As part of this programme, the agency plans to introduce a Smart Healthcare Appointment Management Platform that enables patients to schedule medical appointments, receive appointment reminders, access basic health notices, submit medical assistance requests, and track clinic queue status through a unified web-based portal.

During a preliminary technical review, the agency identified that its current infrastructure is fragmented, lacks centralised monitoring capabilities, and does not meet modern security and operational resilience expectations required for handling healthcare-related information and public-facing services.

The agency therefore requires a secure and operational environment capable of supporting the new platform while maintaining system availability, administrative accountability, and recoverability during operational incidents or system failures.

Your organisation has been engaged as a cyber engineering consultancy to design and implement the required solution.

You have been assigned as the lead Cyber Engineering Analyst responsible for delivering a production-ready environment.

Engineering-First Philosophy

Unlike traditional cybersecurity competitions, no infrastructure, software products, deployment templates, reference architectures, or predefined technology stacks will be provided.

Participants are expected to independently determine suitable operating systems, deployment approaches, security mechanisms, monitoring solutions, backup strategies, and operational controls based on sound engineering judgement and industry practices.

The objective of this challenge is not to assess familiarity with any specific vendor or product, but rather to evaluate the participant's ability to independently design, implement, secure, operate, monitor, troubleshoot, document, and recover a system using practical cyber engineering principles.

COMPETITION DURATION

Participants will be given three (3) weeks to complete the challenge.

During this period, participants may develop and deploy their solution using any environment of their choosing.

This may include cloud infrastructure, local virtualisation platforms, containerised environments, physical hardware, or hybrid deployment approaches.

All provisioning activities, software installation, configuration, security implementation, testing, troubleshooting, operational validation, documentation, and demonstration preparation are the responsibility of the participant.

PROJECT REQUIREMENTS

The Healthcare Agency requires a publicly accessible web-based platform capable of supporting appointment and patient engagement services.

The chosen solution must be deployed on an operating system selected by the participant and must incorporate appropriate security measures to protect system availability, confidentiality, and integrity.

The deployed environment must demonstrate secure administrative access mechanisms and effective user management practices.

Participants are expected to implement controls that minimise the likelihood of unauthorised access while maintaining operational usability for legitimate administrators and users.

The solution should demonstrate evidence of secure baseline configuration practices and system hardening activities appropriate for production-facing services.

The application environment must include a database component capable of storing appointment records, user account information, and service-related data.

Participants are expected to implement suitable database protection measures and demonstrate an understanding of least-privilege access principles, credential security, access management, and secure service configuration.

The agency has also identified a requirement for operational visibility and event awareness.

Participants are therefore required to implement logging, monitoring, or observability capabilities that enable administrators to identify system activity, authentication events, operational issues, and potentially security-relevant incidents.

The technologies and approaches selected to fulfil this requirement are entirely at the discretion of the participant.

In addition, the agency requires assurance that critical services and operational data can be restored in the event of accidental deletion, service failure, or infrastructure disruption.

Participants must therefore design and implement an appropriate backup and recovery approach capable of restoring business-critical functionality and data.

Participants are encouraged to consider maintainability, operational simplicity, and practical recovery procedures throughout the design and implementation process.

DELIVERABLES

At the conclusion of the challenge, participants shall submit a technical package containing sufficient evidence to demonstrate the design, implementation, security, operation, monitoring, and recoverability of their solution.

The submission should include:

- Solution architecture diagram
- Technical implementation documentation
- Security configuration evidence
- Administrative access controls documentation
- Monitoring and logging evidence
- Backup and recovery procedures
- Validation or testing evidence
- Supporting artefacts relevant to the solution

Evidence may include screenshots, configuration files, command outputs, logs, scripts, validation results, diagrams, deployment notes, or any additional material considered relevant by the participant.

Participants should assume that assessors have not previously seen their environment.

Documentation should therefore be sufficiently detailed to allow assessors to understand the participant's engineering decisions, implementation methodology, security considerations, and operational approach.

LIVE DEMONSTRATION

Each participant or team will be allocated thirty (30) minutes for a live demonstration and technical defence session.

During this session, participants will be expected to present their architecture and explain the reasoning behind their deployment decisions, security controls, monitoring implementation, backup strategy, and operational design choices.

Participants must demonstrate that their solution is functional and capable of supporting the stated business requirements.

Assessors may request demonstrations involving:

- Administrative access controls
- User management processes
- Security configurations
- Database protections
- Monitoring and logging functionality
- Backup validation
- Recovery procedures
- Troubleshooting activities

The demonstration should also provide evidence that participants understand the engineering decisions they have made throughout the project lifecycle.

Assessors may ask participants to discuss trade-offs, explain alternative approaches, justify technology selections, and describe how operational or security risks were mitigated.

Participants should be prepared to defend both their technical implementation and the reasoning behind their engineering decisions.

ASSESSMENT PHILOSOPHY

The JURUS programme adopts an engineering-first assessment philosophy.

Assessment will therefore focus on demonstrated capability rather than theoretical knowledge alone.

Participants will be evaluated based on:

- Quality of implementation
- Effectiveness of security controls
- Operational usability and maintainability
- Completeness and clarity of documentation
- Monitoring and visibility capabilities

- Practicality of backup and recovery procedures
- Ability to explain and justify engineering decisions

A more complex solution will not necessarily achieve a higher score. Greater emphasis will be placed on solutions that are appropriate, secure, maintainable, well-documented, operationally practical, and clearly understood by the participant.

The highest-scoring submissions will demonstrate not only technical competence, but also sound engineering judgement, operational discipline, and professional communication skills.

EXPECTED OUTCOME

Upon completion of the challenge, successful participants should demonstrate the capability to independently design, build, secure, monitor, troubleshoot, document, and recover a production-ready system using industry-relevant cyber engineering practices.

This outcome aligns directly with the JURUS Level 1 Analyst competency objective of developing practitioners capable of independently building and securing operational systems while applying structured troubleshooting methodologies and sound operational discipline.